Quiz: retake:-

(C6) $\int \sin(\sqrt{x})\,dx$

Let $u = \sqrt{x}$

$du = \frac{1}{2\sqrt{x}}\,dx$

$2du\sqrt{x} = dx$

$2u\,du = dx$

$= \int \sin(u)\, 2u\,du$

$= 2\int u\sin(u)\,du$

Let

| | |
|---|---|
| $v = u$ | $dv = \sin(u)\,du$ |
| $dv = 1\,du$ | $v = -\cos(u)$ |

$= 2\left[-u\cos(u) - \int -\cos(u)\,du\right.$

$= -2u\cos(u) + \int \cos(u)\,du$

$= -2u\cos(u) + \sin(u) + C$

$= -2\sqrt{x}\cos(\sqrt{x}) + \sin(\sqrt{x}) + C$

$\therefore \int \sin(\sqrt{x})\,dx = -2\sqrt{x}\cos(\sqrt{x}) + \sin(\sqrt{x}) + C$

C8: $\int_0^1 \frac{t}{(t^2+1)^{3/2}}\,dt$

Let $u = t^2+1$

$du = 2t\,dt \Rightarrow \frac{du}{2} = t\,dt$

If $u = t^2+1$ and $t=0 \Rightarrow u=1$

$t=1 \Rightarrow u=2$

$$= \int_1^2 \frac{du}{2(u)^{3/2}}$$

$$= \frac{1}{2}\int_1^2 (u)^{-3/2}\,du$$

$$= \left(\frac{1}{2}\right)\left[\frac{u^{-3/2+1}}{-3/2+1}\right]_1^2$$

$$= \left(\frac{1}{2}\right)\left[\frac{u^{-1/2}}{-1/2}\right]_1^2$$

$$= -1\left[(2)^{-1/2} - (1)^{-1/2}\right]$$

$$= (-1)\left[\frac{1}{\sqrt{2}} - 1\right]$$

$$= 1 - \frac{1}{\sqrt{2}}$$

$$= \frac{\sqrt{2}-1}{\sqrt{2}}$$

$$\therefore \int_0^1 \frac{t}{(t^2+1)^{3/2}}\, dt = \frac{\sqrt{2}-1}{\sqrt{2}}$$

Q. $\int \frac{2x^2 - (1+c+5}{x^3 - x^2 + x - 1}\, dx$